

C-DAC PG-DAC Complete Syllabus

C++ Programming (90 Hrs)

C++ Basics, Tokens, Operators & Expressions, Conditional & Looping Statements, Arrays, Functions, Memory Management & Pointers, Classes & Objects, Constructors & Destructors, Inheritance, Polymorphism, Operator Overloading, Virtual Functions & Abstract Classes, Namespaces, Exception Handling, File Handling, Templates, STL, RTTI, Dynamic Memory Allocation.

Database Technologies (90 Hrs)

Introduction to DBMS & Database Terminologies, Data Models, Database Design & ER Modeling, SQL (DDL, DML, DCL, TCL), Data Redundancy & Anomalies, MySQL Data Types & Constraints, Aggregate Functions & Grouping, Joins & Relational Algebra, Subqueries & Correlated Queries, Views & Indexes, Window Functions, CTE, Transactions & ACID Properties, Stored Procedures, Functions, Flow Control Statements, Cursors, Triggers, Exception Handling, Temporary Tables, NoSQL Concepts, MongoDB, CRUD using MongoDB.

Operating Systems & Software Development Methodologies (60 Hrs)

Operating Systems, Linux & File System, Linux Commands & Permissions, Shell Programming, Regular Expressions, Processes & Threads, Memory Management, Virtual Memory, Deadlocks & Synchronization, I/O Management, Git & Version Control, SDLC, Requirements Engineering, Software Engineering & DevOps, UML, Design Models, Agile & Scrum, Testing Basics, Selenium Automation Testing, Jenkins & Pipeline Integration.

Object Oriented Programming with Java (150 Hrs)

Java Basics & Syntax, JVM & JDK Tools, Data Types, Variables, Operators, Arrays, OOP, Constructors, Static Members, Memory Management, Method Overloading/Overriding, Abstract Classes, Interfaces, Packages, Access Modifiers, Inner Classes, Enums, Lambda Expressions, Functional Interfaces, Garbage Collection, Wrapper Classes, Strings, Exception Handling, File I/O, Serialization, Date & Time API, Collections, Comparable & Comparator, Stream API, Multithreading, Synchronization, Deadlocks, Generics, JDBC.

Algorithms & Data Structures using Java (90 Hrs)

Problem Solving, Computational Thinking, ADTs, Arrays, Stacks, Queues, Circular Queues, Linked Lists, Recursion, Trees, Searching, Sorting, Hashing, Graphs, Algorithm Design Techniques, Performance Analysis, Time & Space Complexity, Big-O, Asymptotic Analysis.

Web Programming Technologies (150 Hrs)

HTTP, HTML, CSS, Tailwind CSS, JavaScript, DOM, JSON, jQuery, TypeScript, AJAX, Axios, Node.js, Async Programming, npm, Express.js, REST APIs, React.js (Components, Props, State, Lifecycle), Event Handling, Conditional Rendering, Forms.

Web-based Java Programming (150 Hrs)

J2EE Architecture, Servlets, Session Management, Request Dispatcher, JSP, MVC, JSTL, Hibernate ORM, Spring Framework, Spring MVC, Spring Boot, Maven, Spring Data JPA, RESTful Web Services, Security, Microservices.

Microsoft .NET Technologies (120 Hrs)

.NET/.NET Core, CLR, Managed Code, C#, OOP, Generics, Collections, Delegates, Events, Lambda Expressions, LINQ, PLINQ, Reflection, File I/O, Multithreading, ASP.NET MVC Core, Razor, ADO.NET, Entity Framework, Web APIs, REST, Security, React Integration, Deployment.

Aptitude (60 Hrs)

Number System, Ratio, Percentage, Profit & Loss, SI/CI, Time-Speed-Distance, Time & Work, Pipes & Cisterns, Boats, Averages, Mixtures, Probability, Permutations & Combinations, Series, Blood Relations, Coding-Decoding, Seating Arrangement, Syllogism, Venn Diagram, Data Interpretation, Ages, Clock & Calendar, Alphabetical Reasoning, Ranking, Direction, Puzzles, Statements & Assumptions.

Effective Communication (60 Hrs)

Personality Development, English Grammar, Correct Usage of English, Listening, Reading, Writing, Formal Applications, Public Speaking, Presentation Skills, Group Discussions, Personal Interviews.

Project (180 Hrs)

End-to-end project including Requirements, Design, Database Modeling, Coding (Java/.NET/Full Stack), Testing, Documentation, Git, Deployment, Final Presentation. Domains: Web Apps, Enterprise Systems, Cloud Solutions, Automation Tools, Data Processing Apps.